FCOM gap fixes (program overview)

Overview of the 207 own commits of the FCOM gap program: A380X and A32NX alerts, displays, FADEC, FMS predictions and VNAV, GPWS / TAWS predictive terrain and obstacles, flight control laws and system models, grouped in feature families with how to test each

Aircraft	A32NX and A380X
Where to find it	Everywhere in the cockpit: E/WD and SD alerts, PFD / ND / MFD / MCDU pages, overhead panels, flyPad Failures
Branch	`program/fcom-gap-fixes` : a fork-only program branch cut from develop `9ad68d612` (it needs develop; it is not an upstream pull request branch), with FlyByWire master 66021c244 merged in. The same changes were placed on single-topic `core/*` branches where their base allows it; this manual describes them as one program. The 207 commits are listed by family in the commit list of the branch.
Status	Fork-only program. Mixed: the 2026-10-12 to 2026-10-14 gap batches were committed without sim tests; the later FADEC, VNAV, A32NX / A380X ECAM, ELAC, A380X F/CTL and GPWS / TAWS work was sim-tested in install groups (per family below). 15 of the commits are `test(...)` , `docs` and `chore` commits (spec conversions, flaky test fixes, README files for the TAWS grid generators, CHANGELOG housekeeping): no user-visible change.
Pictures	Pictures only in the GPWS / TAWS family (in-sim screenshots of the ND TERR AHEAD message and threat areas, taken during development); all other families are described in text.

1. A380X FWS alert units by ATA chapter
2. A32NX FWC alerts, SD and PFD indications
3. A380X SD, PFD, EWD and panel display rules
4. FADEC, EIU, engine idle and relight
5. FMS predictions and VNAV
6. A32NX GPWS: modes 1 and 2 rewritten, terrain ahead alerting (TAD), obstacle alerts and terrain clearance floor (TCF)
7. Surveillance: GPWS test, TCAS traffic and failures, windshear and radio height voices, ILS side
8. Flight control law fixes (ELAC, A380X protections)
9. Aircraft system models: hydraulics, bleed, fuel, electrics, brakes, APU, ice, pressurization, ADIRS
10. Panels, communications, effects, flyPad and GSX

1. A380X FWS alert units by ATA chapter

> **Aircraft:** A380X | **Where:** E/WD, STATUS page, SD pages, overhead panels, flyPad Failures | **Status:** The alerts of 2026-10-28 (avionics ventilation, pack and mixer, hydraulic heat exchanger, CDS DU NOT MONITORED and mailbox access) were sim-tested by the user (DU NOT MONITORED both loops, CAPT / F/O MAILBOX ACCESS FAULT, HYD HSMU, fuel heat exchanger valve, heat exchanger air leak and detection, engine pump overheat): OK. The 2026-10-12 to 2026-10-14 alerts (door, ELEC busbar, landing gear, engine, hydraulic, CAB PRESS, COM, datalink, fuel) were not sim-tested.

What it is for

The A380X flight warning system (FWS) lacked many alerts of the FCOM. They were added unit by unit, by ATA chapter, each with its trigger, flight phase inhibitions, procedure lines, STATUS lines and, where the alert had no way to be triggered, a new flyPad failure: air conditioning (pack valve, pack overheat, pack regulation, mixer, ram air, cabin pressure sensors), avionics ventilation, cabin pressure, doors and T.O CONFIG for every door, ELEC busbars, landing gear and NAV disagree, engines (levels, THR LEVER ABOVE IDLE, attention box, tire pressure), hydraulic pressure switches and heat exchangers, fuel, CDS and mailbox access, datalink and COM, CIDS, APU. It also adds the unit-feet radio height callouts, the PITCH PITCH and COMPANY MESSAGE voices and a fix that keeps the FWS sound manager from looping when the audio functions are lost.

How to use

1. Nothing to set: the alerts appear on the E/WD when their condition occurs.
2. To provoke one: flyPad > Failures, choose the failure named in the test below.

How to test

#	Do this	Expected result
1	A380X powered on the ground or in cruise. flyPad > Failures > ATA 21 > **Pack 1 overheat**, Fail.	AIR PACK 1 OVHT on the E/WD with its procedure; Restore clears it.
2	flyPad > Failures > ATA 21 > **Avionics ventilation left blowing filter clogged**, then the right one.	After both: VENT AVNCS BLOWING DEGRADED on the E/WD (both circuits on their backup valves).
3	flyPad > Failures > ATA 79 > **Engine 1 oil filter clog**.	ENG 1 OIL FILTER CLOGGED comes without chime and without MASTER CAUT (FCOM level).
4	flyPad > Failures > ATA 23 > **CIDS 1 (cabin intercommunication data system)**, then CIDS 2 and 3.	With all three failed: CAB COM CIDS 1+2+3 FAULT.

2. A32NX FWC alerts, SD and PFD indications

> **Aircraft:** A32NX | **Where:** E/WD, STATUS, SD pages, PFD, ND | **Status:** Sim-tested by the user in the coverage program groups G2 / G3 (A32NX ECAM: NAV, F/CTL, mixed alerts, all 6 cards passed, 2026-10-20). The 2026-10-12 to 2026-10-14 alerts (AIR bleed, HYD, ELEC, fuel, doors, oxygen, STATUS paging, SD pages) were not sim-tested.

What it is for

The A32NX flight warning computer got the missing FCOM cautions with their procedure and STATUS lines: AIR bleed cautions, HYD cautions and warnings, ELEC bus, ESS bus and TR alerts, F/CTL surface cautions (both ELACs failed show one ELAC 1+2 FAULT), wing anti-ice, L/G DOORS NOT CLOSED, ventilation, COM, WHEEL HYD SEL FAULT, SIDESTICK and FAC alerts. The SD pages (BLEED, HYD, ELEC, DOOR/OXY, WHEEL, APU) show amber where the FCOM says so; the STATUS page now pages instead of cutting after 16 lines; the PFD gets V1 INOP and ILS / DME flags and the FMA line 3 messages. Pneumatic, electrical and brake control failures were added to the flyPad.

How to use

1. Nothing to set. Provoke an alert with flyPad > Failures.

How to test

#	Do this	Expected result
1	A32NX in cruise. flyPad > Failures > ATA 27, fail **ELAC 1** then **ELAC 2**.	One alert ELAC 1+2 FAULT (not two ELAC FAULT alerts) with its procedure.
2	flyPad > Failures > ATA 36 > **Engine 1 bleed valve stuck closed**, with the engine running and the bleed on.	ENG 1 BLEED FAULT caution with its procedure; the SD BLEED page shows the bleed valve amber when it is not in the commanded position.
3	Cabin altitude above 14,000 ft (rapid decompression failure, flyPad > Failures > ATA 21).	The passenger oxygen masks drop by themselves.
4	Make a STATUS page longer than 16 lines per column (several failures).	A green arrow shows that more follows; CLR shows the next page.

3. A380X SD, PFD, EWD and panel display rules

> **Aircraft:** A380X | **Where:** SD ELEC / F/CTL / WHEEL / BLEED pages, PFD, EWD, glareshield, KCCU | **Status:** Committed in the FCOM gap program of 2026-10-12 to 2026-10-14 (batches P1 to P20 and Tier 2) without sim tests, on the decision that the community tests at pull request time. Unit tests only.

What it is for

Display rules of the A380 FCOM that were wrong or missing: SD ELEC AC / DC (galley and commercial shed legends, generator DISC and OVHT, APU BAT voltage), F/CTL page (BKUP CTL ACTIVE), WHEEL page (WHEEL MORE, BCS / EBCU / SCS lines, brake arcs), automatic SD pages (CRUISE from 1,500 ft after take-off), PFD radio height (median of three RAs, same on both PFDs), the CAPT PFD reading the CAPT side FMS data, FMA messages (MAN DTO for a derated take-off, SPD BRK, CHECK APPR SEL), the VD radio altitude zone, the OANS destination arrow, the AGENT SQUIB light and the loudspeaker volume knobs, and the hosted display gates (a hidden gauge no longer takes the mouse of the DU).

How to use

1. Nothing to set.

How to test

#	Do this	Expected result
1	A380X, take off and climb through 1,500 ft.	The SD shows the CRUISE page from 1,500 ft after take-off (not 60 s after the flaps are up).
2	Lose all PRIMs and SECs (flyPad > Failures > ATA 27) and look at the SD F/CTL page.	BKUP CTL ACTIVE in amber.
3	Final approach below 5,000 ft RA with the VD on the ND.	An amber radio altitude zone under the aircraft symbol; the radio height is the same on both PFDs.

4. FADEC, EIU, engine idle and relight

> **Aircraft:** A32NX and A380X | **Where:** E/WD ENGINE indications, SD ENGINE page, thrust levers, ENG MASTER, flyPad Failures | **Status:** Sim-tested by the user in the coverage program group G1 (2026-10-18, cards A1-A5, B1-B7, F1, F3-F6 passed; the crank-without-FADEC bug it found was fixed). A32NX automatic TOGA after 8 s: sim-tested and OK (2026-10-25). Bleed status fault idle (HI GND IDLE) and A380X pack controllers lost (ENG IDLE HI): sim-tested OK on 2026-10-22. The 2026-10-12 to 2026-10-14 engine batches (P15, HP fuel valve, FADEC faults, derated take-off) were not sim-tested.

What it is for

FADEC power supply and channels (A32NX alternator and DC buses, A380X EEC channels and EIPMs), the engine interface units, idle modes (approach idle, higher idle with packs, HI GND IDLE with an ENG BLEED STATUS FAULT, ENG IDLE HI with both pack controllers lost), the A380X NORMAL / DEGRADED (N1) modes with the degraded accuracy indications, the A32NX automatic TOGA 8 s after the levers are set below TOGA without FLEX, automatic relight ignition, the N3 run-down and quick relight, the A380X engine overspeed failure, the FCOM EGT limits, REV, FLEX and N2 crank indications on the E/WD, and the pre-start motoring of a hot engine.

How to use

1. Nothing to set. Engine and FADEC failures are in flyPad > Failures (ATA 72, 73, 74, 77, 78, 79, 80).

How to test

#	Do this	Expected result
1	A32NX at take-off without a FLEX temperature: set the thrust levers between CL and TOGA and leave them.	ENG THR LEVERS NOT SET; 8 s later the FADEC selects TOGA thrust, until the levers go back to CL or below.
2	A380X on the ground, ENG MASTER 1 OFF at idle, then back ON after 10 s.	The N3 runs down slowly (about 45 % after 10 s); ENG MASTER ON relights the engine while the N3 shown is still above the relight limit.

3	A380X in flight: flyPad > Failures > ATA 73 > **Engine 1 overthrust (thrust control lost)**.	The FADEC shuts engine 1 down without automatic relight and ENG 1 FAIL shows the THRUST CTL LOST line.
4	A380X: fail both pack controllers (flyPad > Failures > ATA 21).	COND TEMP CTL FAULT, engines to idle high at idle thrust, STATUS INFO ENG IDLE HI.

5. FMS predictions and VNAV

> **Aircraft:** A32NX and A380X | **Where:** MCDU F-PLN, PERF, PROG, FUEL PRED, A380X MFD F-PLN, PERF, VERT REV, FUEL&LOAD, ND and VD | **Status:** Sim-tested by the user in the coverage program groups G1, G2 and G6 (2026-10-18 to 2026-10-23): energy circle, FLS beam, CRZ WIND page SAT / ALT, ALTN legs, SEC copy predictions, OPT FL with SAT entry: OK. Go-around and engine-out predictions and the 2026-10-12 / 13 MFD and MCDU page fixes (P8, P20, hold, EOSID, POSITION MONITOR): not sim-tested.

What it is for

The flight plan predictions and the VNAV logic follow the FCOM: ECON speeds and OPT FL from the cost index, weight and wind, the constant Mach segment, RTA time constraints with speed control, optimum step and VNAV messages, the energy circle and DES CABIN RATE, descent predictions with anti-ice, predictions on the alternate and secondary flight plans, go-around, take-off and engine-out predictions with drift down, SAT entries for the cruise, time markers as pseudo waypoints, MORE DRAG and SPD BRK messages, holds (database hold, last exit time, unfolded on the VD), the EOSID, the FM position monitor, FUEL PRED and PREDICTIVE GPS pages of the A32NX MCDU.

How to use

1. Enter the flight plan and the PERF data as usual; the predictions appear on the pages named above.

How to test

#	Do this	Expected result
1	A32NX with a flight plan, ZFW and cost index entered. Open PERF and PROG in cruise.	ECON cruise speed and OPT FL follow the cost index, weight and wind (not a fixed table and M.78).
2	A380X in descent toward the destination, ND in ARC or ROSE NAV.	A dashed energy circle within 180 NM of the destination; the VD shows the FLS beam on a non precision approach.
3	A380X MFD > F-PLN: add a hold with the HOLD revision; open the hold page and press DATABASE.	The hold becomes the database hold; LAST EXIT time and EFOB are predicted.
4	A380X: fail one engine in cruise (flyPad > Failures > ATA 72) and open PERF CRZ.	EO MAX, DRIFT DOWN TO FL and the EO CRZ MODE are shown.

FLS messages: NO FLS FOR THIS APPR and DISCONNECT AP FOR LDG

NO FLS FOR THIS APPR (amber FMS message: A32NX MCDU scratchpad, A380X MFD message area) shows when the approach you insert cannot be flown with the FLS: an LDA, SDF or TACAN approach, or an approach whose final course is more than 50 degrees off the runway. Clear it with CLR (A32NX) or CLEAR INFO (A380X). It comes back for each new approach of that kind. **DISCONNECT AP FOR LDG** on FMA line 3 now also shows on an F-G/S approach, because the FLS beam is not a precision approach. A32NX: autopilot still on at the minimum minus 50 ft, or 400 ft above the ground without a minimum. A380X: at the minimum minus 50 ft, or 350 ft above the landing elevation without a minimum (no longer a fixed 150 ft radio height).

#	Do this	Expected result

1	A32NX or A380X: insert an LDA, SDF or TACAN approach (or one whose final course is more than 50 degrees off the runway) in the destination.	Amber NO FLS FOR THIS APPR. CLR / CLEAR INFO removes it; a new approach of that kind brings it back.
2	F-G/S approach, AP on, descend to the minimum minus 50 ft (400 / 350 ft above the ground without a minimum).	Amber DISCONNECT AP FOR LDG on FMA line 3. Disconnect the AP.

6. A32NX GPWS: modes 1 and 2 rewritten, terrain ahead alerting (TAD), obstacle alerts and terrain clearance floor (TCF)

> **Aircraft:** A32NX | **Where:** Overhead GPWS panel (GPWS SYS, TERR, G/S MODE, FLAP MODE, LDG FLAP 3 pbs), GPWS/PULL UP lights on both glareshields, EMER CANC key, ND terrain display | **Status:** Tested in the sim by the user with the recorder (October 2026): take-off without false alert, FL100 toward Mount Fuji with the caution, the warning and AVOID TERRAIN; the ND TERR AHEAD message and threat areas tested the same way on 26 October 2026, and the AVOID TERRAIN voice the same day; the obstacle alerts at 3000 ft toward a 3125 ft tower near KFAR the same day (caution, warning, AVOID OBSTACLE); the obstacles in the ND terrain picture at Hillsboro (KVLY mast drawn red) on 28 October 2026. Not tested: the Himalaya at FL250, Innsbruck-type valley approaches. Unit tests for every part.

What it is for

Before, the A32NX GPWS only knew the basic modes, which work from the radio altimeter. Flying toward a mountain at FL100 gave no alert until the ground rose under the aircraft, far too late. The GPWS now also looks ahead: it knows the terrain around the aircraft from an elevation grid, and warns when your flight path and the terrain meet. Modes 1 and 2 were written again at the same time.

What you see and hear

Situation	You hear	You see
Terrain ahead, you can still avoid it (caution)	"TERRAIN AHEAD", repeated every 7 s	Amber GPWS light on both GPWS/PULL UP pbs; the terrain comes up on both NDs by itself, with an amber **TERR AHEAD** and the threat cells in yellow
Terrain right below the aircraft, too low for the place (terrain clearance floor)	"TOO LOW TERRAIN" once, and again each time you lose another fifth of your height above the floor	Amber GPWS light for as long as you are under the floor; the terrain does not come up by itself and there is no ND message
You are too low over ground that rises ahead (the look-ahead finds terrain that is lower than the aircraft)	"TOO LOW TERRAIN", repeated every 7 s	Amber GPWS light; the terrain does not come up by itself and there is no ND message
Terrain ahead, close (warning)	"TERRAIN AHEAD" once, then "PULL UP" again and again	Red PULL UP light; terrain on both NDs, red **TERR AHEAD**, the threat cells in red
Terrain ahead and the aircraft could not climb over it even with a pull-up	"AVOID TERRAIN" again and again (about every 1.5 s)	Red PULL UP light; terrain on both NDs, red **TERR AHEAD**, the warning cells in a red and black crosshatch
An obstacle (tower, mast) ahead, you can still avoid it (caution)	"OBSTACLE AHEAD", repeated every 7 s	Amber GPWS light; the terrain comes up on both NDs with an amber **OBST AHEAD** and the threat cells in yellow
An obstacle ahead, close (warning)	"OBSTACLE AHEAD" once, then "PULL UP" again and again	Red PULL UP light; red **OBST AHEAD**, the threat cells in red
An obstacle ahead that a pull-up could not clear	"AVOID OBSTACLE" again and again (about every 1.5 s)	Red PULL UP light; red **OBST AHEAD**, the warning cells in the red and black crosshatch

Mode 1 second boundary (sink rate too high near the ground), or mode 2 warning	"PULL UP"	Red PULL UP light
Any other basic alert (SINK RATE, mode 2 caution TERRAIN, DON'T SINK, TOO LOW ..., GLIDESLOPE)	The matching words	Amber GPWS light

- A warning takes over from a caution. Once AVOID TERRAIN has been said, PULL UP does not come back in the same warning.
- Terrain comes first: when terrain and an obstacle threaten at the same level (both a caution, or both a warning) you get the terrain words. An obstacle warning still beats a terrain caution, and a terrain warning beats an obstacle caution.
- An alert stays on for 2 s after the threat has gone, so that a cell at the edge of the strip cannot make the light and the words flicker. On the ground, and with the TERR pb OFF, there is no alert at all.
- The terrain picture comes up by itself on both NDs (ARC or ROSE). The first time it does so in an alert, the ON light of a TERR ON ND pb that was off comes on, so that you can switch it off again. The picture stays on the NDs for as long as the alert lasts, whatever the pb says; when the alert is over it goes away unless the pb is still on. In PLAN mode and at 160 and 320 NM it cannot be drawn (see the ND messages below).
- **The AVOID TERRAIN voice:** the sound bank has no such word, so "AVOID" was made with a speech synthesiser converted to the GPWS voice and joined to the "TERRAIN" word of the sound bank (`tools/gpws-voice/generate.py` rebuilds it). The GPWS says one word at a time: the last PULL UP ends before the first AVOID TERRAIN, they never overlap. "AVOID OBSTACLE" is made the same way and waits for the last OBSTACLE AHEAD or PULL UP in the same way.
- **TERR STBY:** the predictive functions need the GPS position (accurate to 0.5 NM or better), the GPS altitude, the track and ground speed, the vertical speed and the terrain grid around the aircraft, and ADIRU 1 aligned with a valid position, altitude, heading and vertical speed. When one is missing for 3 s the green memo **TERR STBY** comes up on the E/WD, the terrain picture leaves both NDs, the TERR ON ND pbs are switched off and there are no TAD or TCF alerts (the basic modes stay). 1 s after everything is back the functions work again and the memo goes (press the TERR ON ND pb again if you want the picture). A block of the grid that is merely slow to load does not make it flicker. With the TERR pb OFF the memo is **TERR OFF** instead, and with a failed EGPWC you get the caution **GPWS TERR DET FAULT** (its line asks for GPWS TERR OFF) or **GPWS FAULT**.

On the ND: TERR AHEAD and the threat areas

- **OBST AHEAD** takes the place of TERR AHEAD, in the same colours and with the same flashing, when the alert is about an obstacle. The obstacle cells are drawn in the threat areas like terrain cells.
- **TERR AHEAD** on both NDs: amber for the caution TERRAIN AHEAD, red for the warning (TERRAIN AHEAD PULL UP or AVOID TERRAIN). TOO LOW TERRAIN has no ND message (FCOM: nil). It takes the place of the cyan **TERR** label above the terrain elevation box.
- It flashes for 9 s each time it is triggered (again when the caution becomes a warning), then stays steady while the alert lasts.
- If the terrain picture cannot be shown, the ND says why in its centre, red or amber with the alert: **TERR: CHANGE MODE** in PLAN mode, **TERR: REDUCE RANGE** at 160 and 320 NM.
- **Threat areas** are drawn over the terrain picture: the 1 arcminute cells (about 1 NM) that break the caution envelope in solid yellow, those that break the warning envelope in solid red, and in a red and black crosshatch while AVOID TERRAIN is on.
- They come from the look-ahead strip along your track, which is narrow and widens in turns. So they are small patches where your path meets the terrain, not whole mountains. Up to 64 runs of cells are drawn; when there are more, the nearest ones are kept.



Figure 1. Caution (ARC, 20 NM): amber TERR AHEAD at the lower right, the threat cells in yellow on the track.

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.



Figure 2. Warning: red TERR AHEAD, the warning cells in red next to the yellow caution cells.

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.



Figure 3. AVOID TERRAIN: the warning cells in the red and black crosshatch, yellow caution cells beside them.

In-sim screenshot taken during development: not representative of the final in-game UI, reference for the explanation only.

The functions in plain words

- **Terrain ahead alerting (TAD):** the GPWS scans the terrain along and beside your track, and checks whether you could clear it by levelling off and then climbing gently (caution), and whether you could still clear it if you pulled up now (warning). Both envelopes end on such a climb path, so the caution comes about a minute before the

threat and the warning about half a minute before (the 3 degree climb of the caution path is a design choice; the manual gives no angle). It does not alert on the ground.

- **Obstacle alerts:** the same scan also reads an obstacle grid (the highest obstacle top in each cell) and gives the OBSTACLE words for it. No extra margin is added for obstacles, because their heights are surveyed values, not averaged terrain.
- **Terrain clearance floor (TCF):** around the destination runway and the departure runway (up to 15 NM) there is an invisible floor above the ground, measured with the radio altimeter. It is nothing within a quarter of a mile of the runway, then rises from 30 ft to 400 ft at 4 NM, stays at 400 ft out to 12 NM and reaches 700 ft at 15 NM. If you descend below it, you get TOO LOW TERRAIN whatever the gear and flaps are. Around the destination runway there is a second floor, measured from the GPS altitude above the runway elevation, between 1 and 5 NM out and up to 300 ft, for fields where the radio altitude is misleading. The shape of the floor is rebuilt from the figures of the Honeywell guide; it is not a published curve.
- **How close is too close:** the look-ahead keeps a clearance above the terrain that depends on where you are. It is 700 ft in level flight (500 ft when descending) en route, 350 ft (300 ft) in the terminal area, 150 ft (100 ft) on the approach and 100 ft on departure. The terminal area is within 15 NM of a runway below a line from 3500 ft at 15 NM to 1900 ft at 5 NM above the runway; the approach area is within 5 NM and below 1900 ft; the departure area is within 15 NM of the departure runway and below 1500 ft. Close to the departure runway the clearance grows by 100 ft per NM, so that a normal lift-off does not alert.
- **Mountainous approach areas:** where the terrain within 6 NM of the runway is more than 2000 ft above it, the warning looks less far ahead the nearer you are to the runway: 120 s at 6 NM, down to 30 s at the runway, so that you are not warned on every approach there.
- **Landing tunnel:** when you are configured for landing (gear down, landing flaps), inside the approach sector of the runway (2.7 NM), in a tunnel above the approach path, and your track converges on the runway, the alerts are inhibited. That is how a normal landing near hills stays quiet.
- **Airport surroundings:** within 1.5 NM of the runway, ground that is not more than 400 ft above the runway is taken for the airfield itself and does not alert (the 1 arcminute cells around an airport hold the highest point of their surroundings).
- **Take-off:** no alerts on the ground, and the scan starts afresh at lift-off. When the departure runway is known only as the point where the roll started (no runway in the flight plan) the terrain clearance floor does not work within 2.5 NM of that point.

Which runways the GPWS knows

The A32NX has no airport database in its GPWS. Everything that depends on a runway (the terrain clearance floor, the phase of flight and its clearance, the mountainous approach area, the airport surroundings and the landing tunnel) uses only the runways below, and the one nearest to the aircraft counts.

Runway	Where it comes from	What the GPWS knows about it
Destination	The destination airport of the flight plan (FMGC 1). With a runway or an approach chosen, the runway; without one, only the airport reference point.	The runway threshold (a displaced threshold is allowed for), its direction, length and elevation. For the airport reference point alone, only that point and its elevation.
Departure (flight plan)	The runway chosen for the origin airport in the flight plan.	Its start, direction, length and elevation. Used as the departure runway when the take-off roll starts on it.

Departure (where you stood)	The place where the aircraft last stood or taxied below 35 kt on the ground; if there is none yet (a flight loaded on the runway), the first position on the ground at any speed. It is used when the plan has no origin runway, or when you take off from another runway than the planned one.	That point and its elevation only, no direction and length.

- Without a destination in the flight plan, or after you have taken the destination out, the only runway is the departure one. Enter the destination and, when you know it, its runway or approach: the landing tunnel and the floor then work with the real runway.
- The departure place stays remembered for the rest of the flight. It is replaced only by the next time the aircraft stands or taxies slowly on the ground.
- A runway counts as the one you are on when you are within a tenth of a mile of it, sideways or before its start. That is how the flight-plan runway is kept for a rolling take-off from the taxiway, and a take-off from a parallel runway uses the place where you stood.

The pushbuttons and keys

Control	What it does
TERR pb OFF (overhead)	Inhibits TAD and TCF. Mode 2 is then active again (it is switched off while the predictive functions work).
GPWS SYS pb OFF	Inhibits the basic modes 1 to 5 and their lights. TAD and TCF keep working: they do not need the radio altimeter or the airspeed.
EMER CANC (ECAM control panel)	Silences the words of the predictive alert that is going on (also the repeats), until that alert is over. The lights stay on. After a pause without alert, a new alert speaks again; a caution that turns into a warning without a pause stays silent.
Stall warning	While the stall warning sounds, the FWC silences the GPWS, the words of the predictive alerts too (the lights stay).
TERR ON ND pb (each side)	Shows the terrain and obstacles on that ND (ARC and ROSE). An alert switches the pb on by itself once and puts the picture on both NDs for as long as it lasts.

Modes 1 and 2 after the rewrite

- The lights follow the real format: **red PULL UP only for the second boundary of mode 1 and for a mode 2 warning** (and the hold after mode 2); every other alert is the amber GPWS light.
- Mode 2 has two nested envelopes: the outer gives the caution ("TERRAIN, TERRAIN"), the inner the warning ("PULL UP"). They depend on the radio altitude, its closure rate and the configuration.
- Mode 2 is inhibited while TAD and TCF work. It comes back when they fail, when TERR is OFF, or when the radio altitude and the height above the terrain computed from the grid differ by more than 2000 ft (one of the two is then wrong).
- The glideslope bias of mode 1 is applied only for a valid glideslope deviation. EMER CANC latches the silence until the alert is over.

Obstacles in the ND terrain picture

- With TERR ON ND selected, ARC and ROSE modes also draw the obstacles (towers, masts) from the obstacle file, in the same colours and dot density as the terrain at that height: an obstacle that is higher than you is red, one about at your level is amber, and so on. There is no separate symbol for an obstacle; this is how the A320 FCOM describes the picture.
- United States only: the obstacle file is the FAA Digital Obstacle File. Elsewhere, or without the file, the ND shows terrain only.

- Nothing is drawn in PLAN mode, as for the terrain.
- The peaks box (highest terrain figure) does not include obstacles: its figures come from SimBridge, which does not know them.

The escape manoeuvre

On a PULL UP or AVOID TERRAIN warning: autopilot off, pitch up firmly, TOGA thrust, wings level, and keep climbing until the alert stops (the usual Airbus terrain escape; the GPWS does not fly it for you).

Limitations and why

Limitation	Why	What to do
Obstacle alerts in the United States only.	The obstacle data is the FAA Digital Obstacle File, which lists the obstacles of the United States and its territories. Elsewhere there is no obstacle alert (and no TERR STBY for it: the terrain functions do not need the obstacle file).	Outside the United States, do not rely on it for man-made obstacles.
The obstacle data is a snapshot.	The grid is built once from the FAA file (26 October 2026); obstacles built or removed later are not in it until the file is rebuilt.	-
An obstacle alert can come a few seconds early.	An obstacle fills its whole 1 arcminute cell (about 1 NM), so the scan meets it up to about 1 NM before the real tower.	-
No runway database: only the destination and the departure runway are known.	TCF, the mountainous approach area, the airport surroundings and the landing tunnel only know the runways of the table above. An airport that is neither the destination nor the departure place has no floor, no tunnel and no cut-off. Above all, an approach to such an airport is not recognised as an approach: the look-ahead keeps its en route clearance of 700 ft (unless the destination or the departure place is within 15 NM), the landing tunnel does not inhibit it, and the alert goes on for as long as the ground threatens, down to touchdown. A landing at an airport that is not the flight-plan destination (an alternate you did not enter, or a flight started in the air without a flight plan) can therefore give TOO LOW TERRAIN, TERRAIN AHEAD, PULL UP and AVOID TERRAIN all the way down to touchdown. This was seen in the sim on 6 November 2026 (runway 28 of RJOA, flight started in the air, no destination); the cause is read from the code.	Enter the airport you are going to land at as the destination, with its runway, or set the TERR pb to OFF for that approach.
Valley airports can alert.	The grid cells are 1 arcminute (about 1 NM), so a narrow valley airport such as Innsbruck sits in cells with the heights of the surrounding mountains. A margin that grows with the relief around the cell is added because the source data is averaged and reads peaks too low.	Expect TOO LOW TERRAIN or TERRAIN AHEAD there; set TERR OFF if you must.
Coverage from latitude -84 to 83.	The terrain source has nothing beyond that.	Beyond it: TERR STBY.

The grid is a file, and without it there is TERR STBY.	The grid is `panel/terrain_grid.bin` in the A32NX package folder (about 20 MB). It is too big for the repository, so it lives in the large-files repository (`TCdr/aircraft-large-files`, branch `feat/a32nx-taws-terrain-grid`). An aircraft without it has no predictive terrain alerts and shows TERR STBY.	Install a build that contains the file.
Threat areas only where the terrain picture is shown.	They are drawn over the terrain picture and cut to its area, so there are none in PLAN mode (TERR: CHANGE MODE instead).	Use ARC or ROSE NAV.
Threat areas are 1 arcminute cells.	That is the resolution of the terrain grid (about 1 NM): the patches are blocky and show the cells of the look-ahead strip, not the shape of the mountain.	-
AVOID TERRAIN is a made-up recording.	The sound bank has no such word: "AVOID" is synthesised and converted to the GPWS voice, so it can sound slightly different from the recorded words.	-
Position accuracy of 0.5 NM or better is needed.	Design choice: the accuracy at which the real system deactivates is not published. Half a cell was chosen so that the cell under the aircraft is the cell read.	-

Where the data comes from

The elevation grid is derived from the terrain data of SimBridge (the data behind the ND terrain display), reduced to one byte per 1 arcminute cell: 0 is water, otherwise the highest elevation of the cell in 50 m steps. It is built by `tools/taws-terrain-grid/build_grid.py`. The source is already averaged (Fuji reads about 3650 m, the real summit is 3776 m), which is why the alert logic adds its own margins. The aircraft reads the file from its own package while you fly; SimBridge does not have to run and nothing goes to the internet.

The obstacle grid `panel/obstacle_grid.bin` (about 265 KB, in the repository) has the same format: in each cell the highest obstacle top above sea level, rounded up to 50 m. It is built from the FAA Digital Obstacle File by `tools/taws-obstacle-grid/build_obstacle_grid.py`.

How to use

1. Nothing to set. Fly with the GPS working and a destination in the flight plan; the GPWS SYS and TERR pbs stay in.
2. Set TERR to OFF only to inhibit TAD and TCF (for instance on a deliberately low valley approach). Press EMER CANC to silence a repeating alert once you have taken note of it.

How to test

#	Do this	Expected result
1	A32NX in flight, GPS aligned, TERR and GPWS SYS pbs in. Look at the ND.	No TERR STBY. (Without `terrain_grid.bin` in the panel folder: TERR STBY.)
2	Take off normally from an airport near hills.	No alert on the ground or at lift-off.
3	Cruise at FL100 on a track that meets high ground (for example RJAA to RJOA, toward Mount Fuji).	About a minute before the terrain: "TOO LOW TERRAIN" then "TERRAIN AHEAD" every 7 s, amber GPWS light, terrain on both NDs.

4	Keep the course.	The warning: "TERRAIN AHEAD" once, "PULL UP" repeating, red PULL UP light. When a pull-up could not clear it: "AVOID TERRAIN", about every 1.5 s, never over a PULL UP.
5	Look at both NDs in ARC at 20 NM during the caution, then the warning, then AVOID TERRAIN.	Caution: amber TERR AHEAD in place of the cyan TERR label, flashing 9 s then steady, yellow cells on the track. Warning: TERR AHEAD red and flashing again, red cells. AVOID TERRAIN: the red cells crosshatched red and black.
6	During the alert, select PLAN mode, then ARC at 160 NM.	TERR: CHANGE MODE, then TERR: REDUCE RANGE, in the ND centre, in the colour of the alert; no threat areas.
7	Before the alert, leave both TERR ON ND pbs off. During the caution switch one pb off if it came on.	Both pbs come on when the alert starts (once) and the terrain is on both NDs. After you switch a pb off the terrain stays on the ND until the alert is over, then goes. A pb you leave on keeps the terrain after the alert.
8	Press EMER CANC during the alert.	The words stop and do not repeat for that alert; the lights stay.
9	Turn away and climb.	The alert ends, the lights go off. A new threat speaks again.
10	In the United States, fly at 3000 ft and 260 kt straight at a tall tower (for example from KFAR, DIR TO 4716.7N/09720.5W, a 3125 ft tower) and keep the level.	About 75 s out: "OBSTACLE AHEAD" every 7 s and amber OBST AHEAD; about 30 s out: "OBSTACLE AHEAD", "PULL UP", red OBST AHEAD; then "AVOID OBSTACLE" about every 1.5 s, never over a PULL UP.
11	Climb (AP off, pitch up, TOGA).	The obstacle alert ends.
12	Park the A32NX at Hillsboro, North Dakota (3H4). Power on, ND in ROSE NAV, range 20 NM, TERR ON ND.	A red patch about 9 NM west of the airport: the KVLY mast (about 2060 ft above the ground). Without `obstacle_grid.bin` in the panel folder the ND shows the terrain only. Nothing in PLAN mode.
13	Set the overhead TERR pb OFF in front of terrain.	No TAD or TCF alert; mode 2 works again from the radio altimeter.

7. Surveillance: GPWS test, TCAS traffic and failures, windshear and radio height voices, ILS side

> **Aircraft:** A32NX and A380X | **Where:** Overhead GPWS panel, EFIS CP TRAF pb, ND, MFD SURV page, PFD | **Status:** Committed in the FCOM gap program of 2026-10-12 to 2026-10-14 (batches P1 to P20 and Tier 2) without sim tests, on the decision that the community tests at pull request time. Unit tests only.

What it is for

Smaller surveillance items of the gap program: the A32NX GPWS can be tested on the ground with the PULL UP-GPWS pushbutton; the TRAFFIC selector of the THRT shows the proximate and other intruders on the ND; the A380X EFIS CP TRAF pb shows all the traffic and a TA or RA selects it; the sidestick priority voices (DUAL INPUT, PRIORITY LEFT / RIGHT), the intermediate radio height callouts and the WINDSHEAR voice; the A380X alpha floor inhibition above M 0.5; each PFD reads the ILS receiver of its own side; the MFD SURV status page shows a failed XPDR, TURB and PRED W/S failures and TAWS 1(2) IN USE.

How to use

1. Nothing to set. GPWS test: on the ground press and hold the PULL UP-GPWS test pushbutton.

How to test

#	Do this	Expected result
1	A32NX on the ground. Press the GPWS test pushbutton on the overhead panel.	GLIDESLOPE with the GPWS light, then PULL UP and TERRAIN TERRAIN PULL UP with the PULL UP light.
2	A380X with traffic around, EFIS CP TRAF pb off, then a TA appears.	The ND selects the traffic display by itself; the TRAF pb on shows all the traffic.
3	Move both sidesticks together on the ground with the A32NX powered.	DUAL INPUT voice every 5 s; a pilot holding priority for 1 s gives PRIORITY LEFT or PRIORITY RIGHT.

8. Flight control law fixes (ELAC, A380X protections)

> **Aircraft:** A32NX and A380X | **Where:** Sidestick, PFD speed scale, flyPad Failures (hydraulics) | **Status:** Sim-tested by the user on 2026-10-25: A32NX blue + yellow hydraulics lost keeps normal law, A32NX high speed protection, A380X 60 / 66 degree bank limit: OK. The A380X protection values were OK but the first version overshot to 389 kt with the stick held (A320 gains); the dive fix (VMO / MMO in a dive, 15 degree nose-down limit held) was not recorded as sim-tested.

What it is for

A32NX: high speed protection stops a full forward sidestick at VMO+16 kt / MMO+0.04 (not 380 kt / M0.88); the loss of blue and yellow hydraulics keeps pitch normal law; the sidestick takeover button must be held 40 s. A380X: A380 values for the high speed protection (active from VMO+10 kt, full forward holds VD-10 kt), bank limit 60 degrees with slats or flaps (67 clean), no nose-up limit drop in CONF FULL, direct law after the loss of the three IRs, abnormal attitude law from 30 degrees angle of attack, and the dive fix.

How to use

1. Nothing to set: fly the aircraft.

How to test

#	Do this	Expected result
1	A32NX in cruise. Lose the blue and yellow hydraulic systems (flyPad > Failures > ATA 29, for example the blue and yellow reservoir leaks).	The PFD stays in normal law for pitch (no alternate law message).
2	A32NX at level flight, push the sidestick fully forward and hold.	The speed stops rising at about VMO+16 kt.
3	A380X in a turn with slats extended, bank beyond 60 degrees.	The bank angle is limited to 60 degrees.

9. Aircraft system models: hydraulics, bleed, fuel, electrics, brakes, APU, ice, pressurization, ADIRS

> **Aircraft:** A32NX and A380X | **Where:** Overhead panels, SD pages, flyPad Failures | **Status:** Committed in the FCOM gap program of 2026-10-12 to 2026-10-14 (batches P1 to P20 and Tier 2) without sim tests, on the decision that the community tests at pull request time. Unit tests only.

What it is for

System models corrected or added from the FCOM gap audit, many with new flyPad failures: A380X brake pressures (5,000 psi system), ROP and RTO autobrake, gear speed and hydraulic supply (GREEN and YELLOW needed), hydraulic reservoir and electric pump modes, bleed overpressure valve, HP ground air, leak detection and crossbleed start; A380X fuel (load alleviation, AUTO GND XFR, trim tank feed, collector cells, XFR pushbuttons in AUTO); A32NX BSCU / ABCU, fuel temperature, wing tank overflow, APU APS3200 limits and automatic shutdowns, DC BUS 2 takeover after 5 s, DC ESS supply, RAT size, computer self-tests after a power outage, wing and engine anti-ice valves, cabin pressure without a landing elevation, ADIRS alignment limits, ADR / IR failures and power supplies, A380X electrical, braking and steering and ADR airspeed / AOA failures.

How to use

1. Nothing to set. The new failures are in flyPad > Failures under their ATA chapter.

How to test

#	Do this	Expected result
1	A380X on the ground with one electric hydraulic pump only.	The pressure rises steadily to normal (no saw-tooth) while the drooped flight controls re-centre.
2	A380X: fail the GREEN hydraulic system in flight, then lower the gear.	The main gears do not extend normally (GREEN and YELLOW are both needed; the nose gear needs GREEN).
3	A32NX in flight: flyPad > Failures > ATA 24 > **DC 1**, Fail.	DC BUS 2 takes over DC BUS 1 and the DC BAT BUS after 5 s.
4	A32NX cold and dark, APU running: cause an APU ECB failure (flyPad > Failures > ATA 49).	The APU shuts down automatically with a FAULT.

10. Panels, communications, effects, flyPad and GSX

> **Aircraft:** A32NX and A380X | **Where:** RMP, CALLS panel, overhead panels, flyPad Ground and Settings, outside view | **Status:** Committed in the FCOM gap program of 2026-10-12 to 2026-10-14 (batches P1 to P20 and Tier 2) without sim tests, on the decision that the community tests at pull request time. Unit tests only. The GSX power sync re-read was part of the GSX power sync work (2026-09-30).

What it is for

Cockpit panel and flyPad items of the program: the A380X STBY RAD NAV key, held PA key, RMP message queue, CALLS panel (each button calls its own station) and ELT selector, the loudspeaker volume knobs, the A32NX CVR automatic power and CVR TEST, COMPANY COM inbox with the received AOC messages, engine and APU fire, engine damage and oil leak effects on both aircraft (Settings > Sim Options > Situation Visual Effects), the ground jettison drip and contrails per engine, the visible A32NX RAT, the exhaust haze per engine, the A380 taxi width check and taxi route with real ATC clearances, the A380 Services door buttons, pushback map performance, external power following the sim ground power and the GSX power sync setting.

How to use

1. Effects: flyPad > Settings > Sim Options, switch **Situation Visual Effects** on or off (on by default).
2. The other items need no setting.

How to test

#	Do this	Expected result

1	A32NX, engine 1 fire (flyPad > Failures > ATA 26 > **Engine 1 fire**), outside view.	Flames at the exhaust of engine 1 until the fire is out (nothing during a FIRE TEST).
2	A380X, CALLS panel: press PILOT REST.	The cabin does not answer CABIN READY; the EMER ON and CALL lights come on steady.
3	A32NX cold and dark, battery on: wait, then start an engine.	The CVR runs by itself during the first 5 min, with an engine running and in flight.